Supplementary Information for
An attractor account of numerosity perception

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This document accompanies the article *An attractor account of numerosity perception*. Supplementary Material 1 details the trial-level fitting; Supplementary Material 2 reports the full model comparison and the constrained cross-duration fits; Supplementary Material 3 the parameter recovery; Supplementary Material 4 the resolution-limit prediction and its robustness; and Supplementary Material 5 the exploratory developmental-dyscalculia reanalysis.

Supplementary Material 1: Trial-level fitting

Data. We reconstruct one row per trial from the openly available data of Cheyette and Piantadosi (2020), grouping by participant and trial and keeping the presented numerosity, the typed response, and the response time (converted to seconds). We use the five exposure durations (40, 80, 160, 320, 640 ms), true numerosities $n = 1$–15, giving $\approx$10,600 trials per duration and $\approx$53,000 in total. No trials are aggregated: the likelihood is evaluated over the individual integer reports.

Response likelihood. For each model and duration the response probability of Eq. (4) marginalises the readout over the sensory channel $y_0 \sim \mathcal{N}(n, \sigma(n))$. The integral is computed *deterministically*, so the objective is free of Monte-Carlo noise. Gauss–Hermite quadrature is inappropriate here: for the attractor models the clean-up map $y_0 \mapsto x^*$ is near-discontinuous (evidence snaps to basins), and Gauss–Hermite carries a 1.7–4% error even at 241 nodes. We therefore use a uniform standard-normal grid with Gaussian weights for *every* model, so the comparison is on identical footing. Against high-sample Monte-Carlo estimates, a grid of 801 nodes matches the integral to $\leq 2 \times 10^{-3}$ (mean 1.5×10^{-4}). The optimiser uses a cheaper grid (201 nodes); the reported log-likelihoods are recomputed at the fitted parameters on a fine grid (1601 nodes).

Lapse channel. Typed responses run up to 99, and a few per cent of guesses fall outside the integer-well support (3.9–6.1% across durations). We never discard them: the final response probability is $p_{\text{final}}(r \mid n) = (1 - \ell)\, p(r \mid n) + \ell \,/\, R_{\max}$ with $R_{\max} = 99$, a uniform contamination that places positive mass everywhere. The lapse is counted in every AIC and BIC. It is fitted separately at each duration in the attractor, static-readout, Weber/ANS, and separate C&P fits. The published-form joint C&P fit instead shares one lapse across durations; we therefore also fit a matched joint C&P variant with duration-specific lapses.

Response-time channel. Response time is modelled as log-normal, $\log \mathrm{rt} \sim \mathcal{N}(\log(t_0 + m(n)), \sigma_{\mathrm{rt}})$, with the task-fixed base $t_0 = 1.30$ s. The metric $m(n)$ is the expected posterior entropy ($\kappa\, \mathbb{E}[H]$, available to every model), the expected convergence time ($\tau\, \mathbb{E}[T]$, attractor only), or their sum (the hybrid; Eq. 3). The RT parameters are fitted to response times with the landscape held fixed at its response-fit values, providing a constrained cross-channel test of the fitted landscape.

Optimisation. The deterministic objective is minimised by `scipy` differential evolution (Storn and Price 1997) followed by a Nelder–Mead polish, with fixed seeds. The attractor models require a larger optimiser budget than the smooth static readouts (a low-budget fit understates them); re-fitting at a higher budget with a different seed changes the reported log-likelihoods by less than one part in 10^3, so the fits are converged and the model ordering is not an optimisation artefact.

Supplementary Material 2: Model comparison and constrained fits

Table S1 consolidates the joint and separate response-model fits and states the sharing configuration used for every row. In the joint power-law attractor and static fits, only the landscape parameters (σ_c, p) are shared; in the constant-width ablations, $p = 0$ is fixed and only σ_c is shared. The sensory parameters (s_0, λ_0) and lapse are duration-specific. The Weber/ANS model has no separable landscape, so its four parameters remain duration-specific in both comparisons. The separate analysis fits all free parameters independently at each duration.

The normative benchmark is the bounded-optimal model of Cheyette and Piantadosi (2020), reimplemented from their published code: a Zipf $(1/n^2)$ prior read out through an

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information-bottleneck posterior whose information budget is $\min(B, Rt)$ bits, with t the exposure duration. Its published joint form shares B, R, and one lapse across durations ($k = 3$). Because the other joint models use duration-specific lapses, the matched C&P comparison shares only B and R and fits five lapses ($k = 7$); this is the C&P row used in main-text Fig. 2a. The separate C&P analysis frees both the information budget and lapse at each duration ($k = 10$), removing the $\min(B, Rt)$ rate-law constraint. Our implementation reproduces the published parameters on simulated data and matches the reference likelihood to machine precision.

Turning the clean-up dynamics on is a zero-parameter change because the dynamics are deterministic given the landscape. The attractor improves AIC by 1326.7 in the separate analysis and by 640.6 in the joint analysis. Per duration, clean-up helps strongly at 40–160 ms and is roughly neutral at 320–640 ms (main-text Fig. 2b). Fixing $p = 0$, and hence $\sigma_n = \sigma_c$, worsens AIC by 6636.4 in the separate analysis and by 3405.3 in the joint analysis. The matched constant-width static model performs worse still. Clean-up therefore remains beneficial under the ablation, but it cannot compensate for removing magnitude-dependent broadening.

The width exponent p is a free parameter of the model (Table S2). Its point estimates grow the wells from narrow and well-separated at small n to wide and overlapping at large n, but the exponent and the scale σ_c trade off strongly and are not identifiable singly. We therefore summarize their joint effect with the continuous width-based index n_{width}, defined by $\sigma_n = 1$ (Supplementary Material 3). What separates our account from the Weber/ANS magnitude code is not a particular width exponent or this proxy, but the discrete pattern-completion architecture: at the identical trial-level convention the attractor beats the magnitude code by $\Delta\text{AIC} = 4270$ (Table S1).

The primary comparison shares one landscape across durations. The attractor still beats the fully duration-specific Weber/ANS fit by $\Delta\text{AIC} = 756.6$ ($\Delta\text{BIC} = 783.3$). It beats the matched C&P fit by $\Delta\text{AIC} = 11{,}372.0$ ($\Delta\text{BIC} = 11{,}283.3$). Allowing C&P a separate lapse at each duration improves its AIC by 653.7 relative to its original shared-lapse form, but does not alter the model ordering. Forcing the attractor landscape to be duration-independent costs $\Delta\text{AIC} = 3513.2$ relative to the separate fits ($\Delta\text{BIC} = 3482.4$). The duration-specific fits therefore reveal an exposure-dependent gain in effective resolution, represented here by narrower fitted wells. A driven-attractor formulation would be required to derive this temporal change within a trial.

Supplementary Material 3: Parameter recovery

To confirm that the fitted quantities are identifiable at the trial level, we ran a parametric recovery: for each duration we generated 20 synthetic datasets from parameters jittered around the fitted solution, sampled with the empirical per-numerosity trial counts and the lapse channel, and refitted them with the exact headline pipeline (100 refits in total). Table S3 reports the recovery of each parameter and of the derived quantities the paper interprets; Figure S1 shows the scatter and the error-correlation structure.

The landscape scale σ_c and exponent p trade off strongly (error correlation $\rho = -0.97$) and are not identifiable on their own, as do the two sensory parameters s_0 and λ_0 ($\rho = -0.98$). The data constrain the combined width profile rather than its scale and slope independently. We therefore interpret the combined sensory-noise profile $\sigma(n) = \sqrt{s_{00}^2 + s_0 n^{\lambda_0}}$ ($\sigma(n = 5)$, $R^2 = 0.99$), the fitted width σ_3 ($R^2 = 0.89$), the lapse ($R^2 = 0.96$), and the continuous index n_{width}, which recovers in rank (Spearman $\rho = 0.98$, median error 0.04 numerosities). Since $n_{\text{width}} = \sigma_c^{-1/p}$ is a heavy-tailed transform of the traded-off pair, a handful of ridge fits admit a low-exponent solution of near-equal likelihood that inflates the pointwise estimate. The index is robust for ordinal comparisons of the fitted width profile, but it is not an estimate of where attractors disappear.

Supplementary Material 4: Width-based and topological resolution limits

We report two resolution indices with different meanings. The continuous width-based index is defined by $\sigma_n = 1$, where the fitted Gaussian-component width equals the unit spacing between integers. Solving $\sigma_n = \sigma_c n^p = 1$ gives $n_{\text{width}} = \sigma_c^{-1/p}$. This definition is equivalent to the heuristic $d' = 1$ (Macmillan and Creelman 2004). It summarizes the fitted width law and does not test whether an integer remains a local attractor.

The topological index is computed from the fitted total potential, $V(x) = -\log \sum_n \exp(g_n(x))$, over all 19 wells used by the response model. For each fit, we evaluated $\partial V/\partial x$ on a dense grid over $x \in [1, 19]$, bracketed every sign change, and refined each root with Brent's method. The sign of $\partial^2 V/\partial x^2$ classified each stationary point as a local minimum or maximum. We assigned each minimum injectively to its nearest integer. For adjacent assigned minima, the barrier height was $V(x_{\max}) - \max\{V(x_{\min,L}), V(x_{\min,R})\}$, where $x_{\max}$ is the intervening maximum. We then defined n_{topo} as the largest consecutive integer from one that retained its own minimum and a positive barrier from the preceding basin.

The two indices differed in every fit (Table S4; main-text Fig. 2e). The duration-specific n_{width} values increased monotonically from 3.48 to 9.77, whereas n_{topo} increased from 2 to 7 and then remained at 7 between 320 and 640 ms. The joint landscape gave $n_{\text{width}} = 8.32$ and $n_{\text{topo}} = 6$. Their numerical difference is expected because n_{width} summarizes component width, whereas n_{topo} tests basin structure directly.

We repeated the complete stationary-point analysis with 5,000, 20,000, and 80,000 grid points before root refinement. All six fits retained identical stationary-point classifications, integer assignments, and n_{topo} values at every resolution. The largest displacement of any refined root was 2.4×10^{-14} units of the number line. The reported topological limits are therefore insensitive to grid resolution.

Because n_{width} falls inside the fitted window, it could in principle be shaped by the large-n tail rather than by the small-number range itself. It is not: refitting on $n \leq 10$ alone returns nearly the same width exponent and n_{width} as the

Table S1. Trial-level response-model comparison under joint and separate fits. k is the total number of free parameters over all five durations. Information criteria use all $52{,}995$ responses and are reported relative to the power-law attractor within each analysis. The joint power-law models share σ_c and p; the constant-width ablations fix $p = 0$ and share only σ_c. In both cases, s_0, λ_0, and lapse are fitted per duration. Weber/ANS remains fully duration-specific. The joint C&P model shares B and R but uses a duration-specific lapse ($k = 7$). In the separate analysis, every free parameter is fitted per duration; C&P has a free information budget and lapse at each duration. Joint-fit BIC uses the pooled sample size, whereas separate-fit BIC is summed over the five duration-specific fits.

Model	k, joint fit	ΔAIC / ΔBIC	k, separate fits	ΔAIC / ΔBIC
Power-law attractor	**17**	**0 / 0**	**25**	**0 / 0**
Power-law readout (static)	17	641 / 641	25	1327 / 1327
Constant-width attractor ($p = 0$)	16	3405 / 3396	20	6636 / 6600
Constant-width readout (static; $p = 0$)	16	3637 / 3628	20	6864 / 6827
Weber/ANS	20	757 / 783	20	4270 / 4233
Normative (C&P 2020)	7	11372 / 11283	10	13243 / 13134

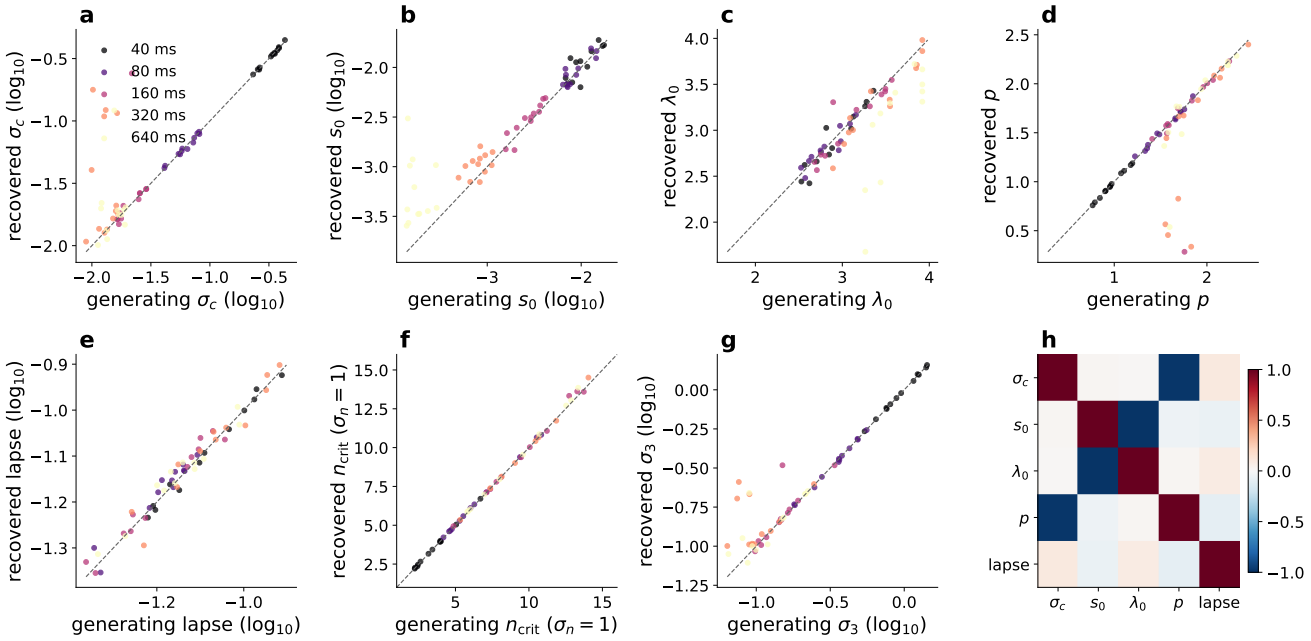


Figure S1. Parameter recovery of the winning model. Recovered versus generating values for the five fitted parameters and the derived n_{width} and σ_3, coloured by exposure duration; dashed lines are identity. The n_{width} panel is clipped to its meaningful range: the on-diagonal bulk recovers well, while a few ridge fits (low exponent, near-equal likelihood) fall off-scale. The bottom-right panel is the Spearman correlation matrix of recovery errors, showing the two strong trade-offs, σ_c–p and s_0–λ_0.

Table S2. Fitted parameters of the winning model across exposure durations. σ_c and p set the well-width law $\sigma_n = \sigma_c\, n^p$; s_0 and λ_0 the sensory-noise profile $\sigma(n)$; ℓ the lapse. The 40 ms flash is noise-dominated (large s_0); from 80 ms the parameters vary smoothly. The scale σ_c and exponent p trade off (Supplementary Material 3) and are not interpreted singly.

Duration	σ_c	s_0	λ_0	p	ℓ
40 ms	0.308	0.011	3.06	0.94	0.084
80 ms	0.057	0.0092	2.81	1.50	0.060
160 ms	0.022	0.0026	3.16	1.80	0.064
320 ms	0.013	0.00077	3.53	1.99	0.081
640 ms	0.015	0.00022	3.98	1.85	0.067

Table S3. Trial-level parameter recovery of the winning model. Identity R^2 and Spearman ρ between generating and recovered values across the 100 synthetic datasets. Scale parameters are compared on the log scale. The interpreted quantities are the derived ones (lower block): n_{width} is a heavy-tailed transform of the strongly traded-off (σ_c, p), so its identity R^2 is uninformative and it is summarised by rank.

Quantity	R^2	Spearman ρ
σ_c (well scale)	0.66	0.76
p (width exponent)	0.15	0.81
s_0 (sensory scale)	0.82	0.94
λ_0 (sensory exponent)	0.41	0.83
ℓ (lapse)	0.96	0.98
n_{width} (derived)	—	0.98
σ_3 (well width at $n = 3$)	0.89	0.89
$\sigma(n = 5)$ (sensory sd)	0.99	1.00

full-range fit at every duration (e.g. $p = 1.67$ vs. 1.50 at 80 ms, 1.78 vs. 1.99 at 320 ms; n_{width} within 0.4 throughout; Fig. S2). Within $n \leq 10$ the attractor still comes out ahead of its static twin ($\Delta\mathrm{AIC} = 271$) and of the Weber/ANS code (1900). This refit establishes that the width-based index is not an artefact of fitting the large-number tail; it is separate from the grid-resolution check of n_{topo} above.

Table S4. Width-based and topological resolution indices. The primary joint fit shares one landscape across durations; the remaining rows are the secondary duration-specific fits. n_{width} is the continuous index defined by $\sigma_n = 1$, whereas n_{topo} is the last consecutive integer retaining its own basin.

Fit	n_{width} ($\sigma_n = 1$)	n_{topo}
Joint	8.32	6
40 ms	3.48	2
80 ms	6.71	4
160 ms	8.32	6
320 ms	9.05	7
640 ms	9.77	7

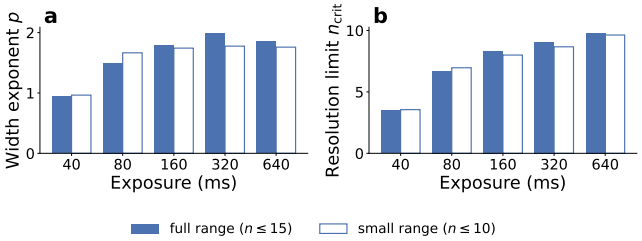


Figure S2. Small-range refit. The fitted width exponent p (**a**) and the continuous width-based index n_{width} (**b**), from the full-range fit ($n \leq 15$, filled) and from a refit on the small range ($n \leq 10$, open). Both are essentially unchanged at every duration.

The response-time channel provides a descriptive comparison around n_{width}. The model tracks the geometric-mean response time over the lower-numerosity values, while its monotone deliberation-time prediction keeps rising where the empirical RT plateaus (Fig. S3). We show solid segments for $n < n_{\text{width}}$ and dashed segments above it to visualize this mismatch, not to define a model-validity or exact-range boundary. The latent-to-seconds RT link is fitted on all n.

The predicted proportion of exact responses shows a small shoulder near n_{width}, even though the fitted width law and sensory-noise profile vary smoothly with n (main-text Fig. 3b). The shoulder vanishes when clean-up is switched off at identical parameters (Fig. S4), so it reflects the dynamics on the integer lattice. Its numerical alignment with n_{width} does not make the width proxy the point where basins disappear; that question is measured by n_{topo}.

Last, we asked whether the fitted model also reproduces the effective Weber fraction. We computed the coefficient of variation (CoV) of the fitted response distribution, $(1-\ell)\,p(r \mid n) + \ell/R_{\max}$, per numerosity and exposure (Fig. S5). The CoV is elevated at the smallest numerosities, where the small denominator inflates $\mathrm{sd}/\mathrm{mean}$ in both model and data. It settles to a roughly constant plateau at larger numerosities. For continuity with the original analysis, we fit $w_{\text{eff}} \propto n^{\gamma}$ over the pre-specified window $n \geq n_{\text{width}}$. This gives a pooled exponent $\gamma = -0.18$ for the model, matching the -0.16 measured in the data. Here n_{width} defines an analysis window, not the onset of an estimation regime.

Supplementary Material 5: Exploratory reanalysis of developmental dyscalculia

As an exploratory extension we applied the winning model to the exact-enumeration data of Decarli et al. (2020): accuracy and response time per subject and numerosity for 32 children with developmental dyscalculia (DD) and 32 age-matched controls, using numerosities $n = 1$–7 (dropping $n = 8$ exactly as the original authors do, for the guessing end-effect they report). The dot arrays were shown for a fixed exposure of 500 ms, so all quantities in this section, including n_{width}, refer to that exposure. The model fits both groups well (Fig. S6a,b) and again reproduces the widening well profile of the main dataset. The fitted $n_{\text{width}} \approx 8$ falls slightly below the adult values of 9.1–9.8 at 320–640 ms (Supplementary Material 4). This is a comparison of fitted width profiles, not of topological basin counts.

First, the decline of accuracy with numerosity is better explained by changing the fitted well-width profile than by sensory noise. With an exhaustive per-subject grid search and a fixed common lapse and exponent, we fitted two single-mechanism variants to each child's accuracy curve. In the width-only variant, sensory noise is switched off and the decline is carried by broader, more overlapping wells. In the noise-only variant, the width profile is fixed and sensory noise carries the decline. The width-only variant fits better in $64/64$ subjects (mean SSE 0.010 vs. 0.097, Wilcoxon $p = 3.5 \times 10^{-12}$; Fig. S6c). This comparison localizes the fitted difference to the width profile; it does not establish where attractors disappear.

Second, the group difference falls on the same width parameter. Per subject, n_{width} is lower in DD (8.1 vs. 8.8 for controls; Mann–Whitney $p = 0.018$, $r_{\text{rb}} = -0.35$ with the RT channel included; $p = 0.044$, $r_{\text{rb}} = -0.29$ in the accuracy-only fits of Fig. S6d), whereas the mirror analysis that frees sensory noise instead shows only a weak trend ($p = 0.05$) and fits worse. In rank terms, the n_{width} difference is comparable to the raw accuracy deficit at $n = 4$–6 ($r_{\text{rb}} = -0.33$). This result concerns the continuous width proxy and does not imply a group difference in n_{topo}.

On this model, then, developmental dyscalculia is modelled as a coarser exact code: the discrete resolution runs out earlier, at the subitizing edge. It is also complementary to the account of Dolfi et al. (2024), who locate the dyscalculic deficit in a weaker number sense within an image-computable network. Both accounts place the deficit in the number code, and coupling the attractor to such a front end (which the present model sets aside) would allow a resolution deficit and a sensory-noise deficit to be separated within a single model. The analysis remains exploratory: the data are coarse (7 trials per cell), the localisation to resolution rests on the per-subject model comparison rather than on a full identification of the two mechanisms, and a small sensory-noise component may coexist.

References

Cheyette SJ and Piantadosi ST (2020) A unified account of numerosity perception. *Nature Human Behaviour* 4(12): 1265–1272. DOI:10.1038/s41562-020-00946-0.

Decarli G, Paris E, Tencati C, Nardelli C, Vescovi M, Surian L and Piazza M (2020) Impaired large numerosity estimation and intact subitizing in developmental dyscalculia. *PLOS ONE* 15(12): e0244578. DOI:10.1371/journal.pone.0244578.

Dolfi S, Decarli G, Lunardon M, De Filippo De Grazia M, Gerola S, Lanfranchi S, Cossu G, Sella F, Testolin A

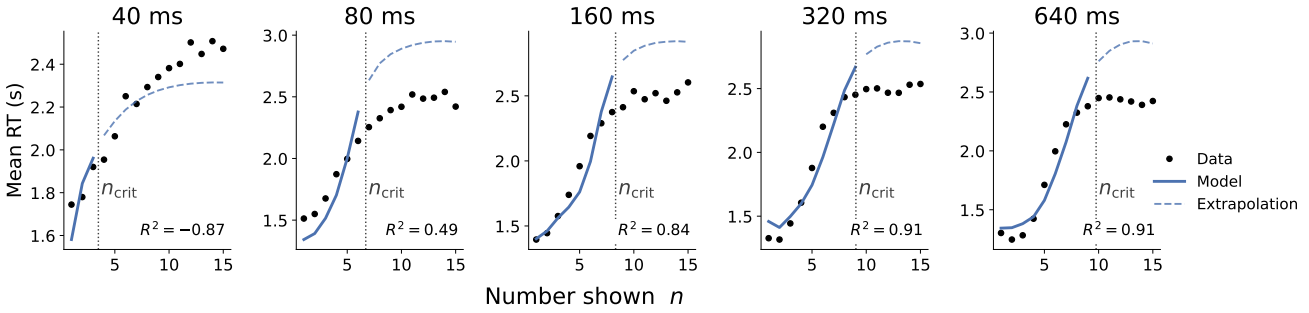


Figure S3. Response time around the width-based index. Predicted geometric-mean RT against the data, shown solid for $n < n_{\mathrm{width}}$ and dashed above it. This segmentation visualizes the location of the continuous width proxy; it does not define an exact-range or model-validity boundary.

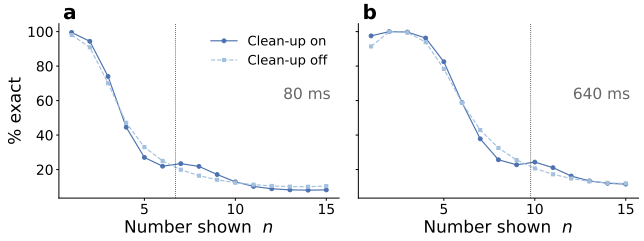


Figure S4. The exact-accuracy shoulder is a clean-up signature. Predicted proportion of exact responses at 80 ms (**a**) and 640 ms (**b**) with clean-up on (solid) and off (dashed), at identical parameters; dotted verticals mark n_{width}. The shoulder occurs near this width-based proxy and disappears without clean-up.

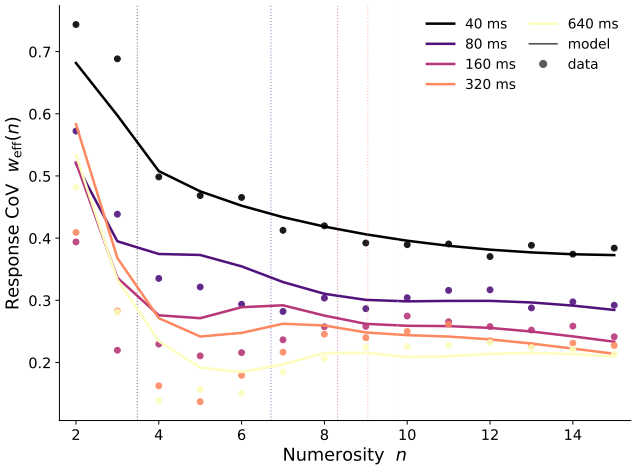


Figure S5. The effective Weber fraction. Coefficient of variation of the response distribution $w_{\mathrm{eff}}(n)$ per numerosity and exposure duration. Solid lines are the model (lapse included, restricted to in-support responses and renormalised), dots are the data, and dotted verticals mark n_{width}. The lines approach a roughly constant plateau at larger numerosities. The verticals identify the width-based analysis cutoff, not a topological or exact-range boundary.

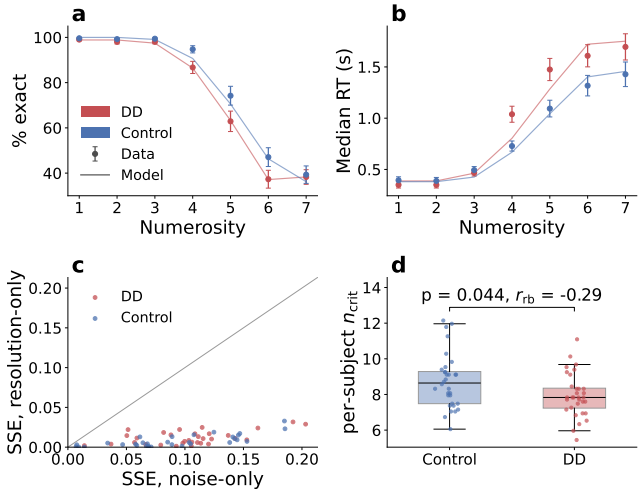


Figure S6. Developmental dyscalculia and the fitted width profile. (a) Exact accuracy per numerosity (points, group mean ± SEM; lines, the group-level model fit). (b) Median response time, same conventions. (c) Per-subject accuracy SSE of the two single-mechanism fits: the width-only model fits better than the noise-only model in $64/64$ subjects (all points below the diagonal). (d) Per-subject n_{width}, defined by $\sigma_n = 1$, from the width-only accuracy fits at the task's 500 ms exposure. Boxes show the median and interquartile range; whiskers show $1.5\times$ IQR; points are individual subjects. The bracket gives the Mann–Whitney p and rank-biserial correlation r_{rb} for this accuracy-only analysis.

and Zorzi M (2024) Weaker number sense accounts for impaired numerosity perception in dyscalculia: Behavioral and computational evidence. *Developmental Science* 27(6): e13538. DOI:10.1111/desc.13538.

Macmillan NA and Creelman CD (2004) *Detection Theory: A User's Guide*. 2 edition. New York: Psychology Press. ISBN 978-1-4106-1114-7. DOI:10.4324/9781410611147.

Storn R and Price K (1997) Differential Evolution - A Simple and Efficient Heuristic for Global Optimization over Continuous Spaces. *Journal of Global Optimization* 11: 341–359. DOI: 10.1023/A:1008202821328.